



The Elements of Data

AWS Academy Data Engineering

Introduction

The Elements of Data

Module objectives

This module prepares you to do the following:

- List the five Vs of data.
- Describe the impact of volume and velocity on your data pipeline.
- Compare and contrast structured, semistructured, and unstructured data types.
- Identify data sources that are commonly used to feed data pipelines.
- Pose questions about data to assess its veracity.
- Suggest methods to improve the veracity of data in your pipeline.

Module overview

Presentation sections

- The five Vs of data – volume, velocity, variety, veracity, and value
- Volume and velocity
- Variety – data types
- Variety – data sources
- Veracity and value
- Activities to improve veracity and value

Activity

- Planning Your Pipeline

Knowledge checks

- Online knowledge check
- Sample exam question

Common data pipeline questions



Data
engineer

- Does the organization have the data that addresses the need? Where is the data stored and in what format?
- Will I need to combine data from multiple sources?
- What is the type and quality of data? What will be the source of truth?
- What are the security requirements for this data? Who needs access to it and in what state?
- What types of mechanisms are needed to transfer the data from its locations into the pipeline?
- How much data is there, and how frequently is it updated or processed?
- How important is speed when data is requested?



- What can the data tell me?
- How will I evaluate the results?
- What kind of visualization do I need?
- Which formats and tools are the analysts familiar with?
- Do I need a big data framework?
- Which type of AI/ML models fit?
- What is the simplest way to implement AI/ML?



Data
scientist

The five Vs of data – volume, velocity, variety, veracity, and value

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Data characteristics that drive infrastructure decisions



Volume

How big is the dataset? How much new data is generated?

Velocity

How frequently is new data generated and ingested?

Variety

What types and formats? How many different sources does the data come from?

Veracity

How accurate, precise, and trusted is the data?

Value

What insights can be pulled from the data?

Strategies that support getting the best value from data

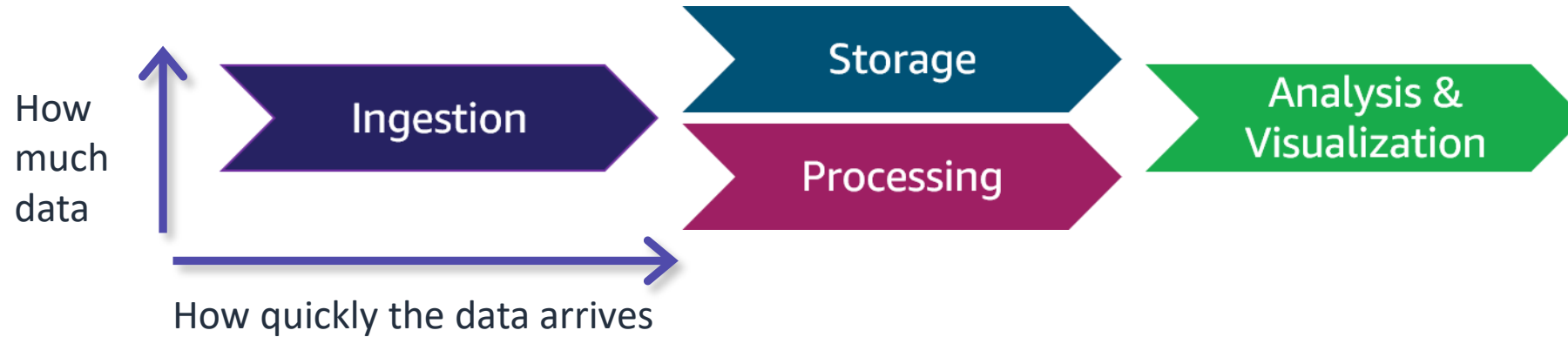
- Confirm available data meets need.
- Evaluate feasibility of acquiring data.
- Match pipeline design to data.
- Balance throughput and cost.
- Let users focus on business.
- Catalog data and metadata.
- Implement governance.



Volume and velocity

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Scaling your pipeline for volume and velocity of data



- The amount of data plus the pace of data drive design choices.
- Volume and velocity impact all layers of the pipeline.
- Evaluate each pipeline layer for its own requirements.
- Balance costs for throughput and storage against the required time to answer and the accuracy of the answer.

Ingestion decisions related to volume and velocity



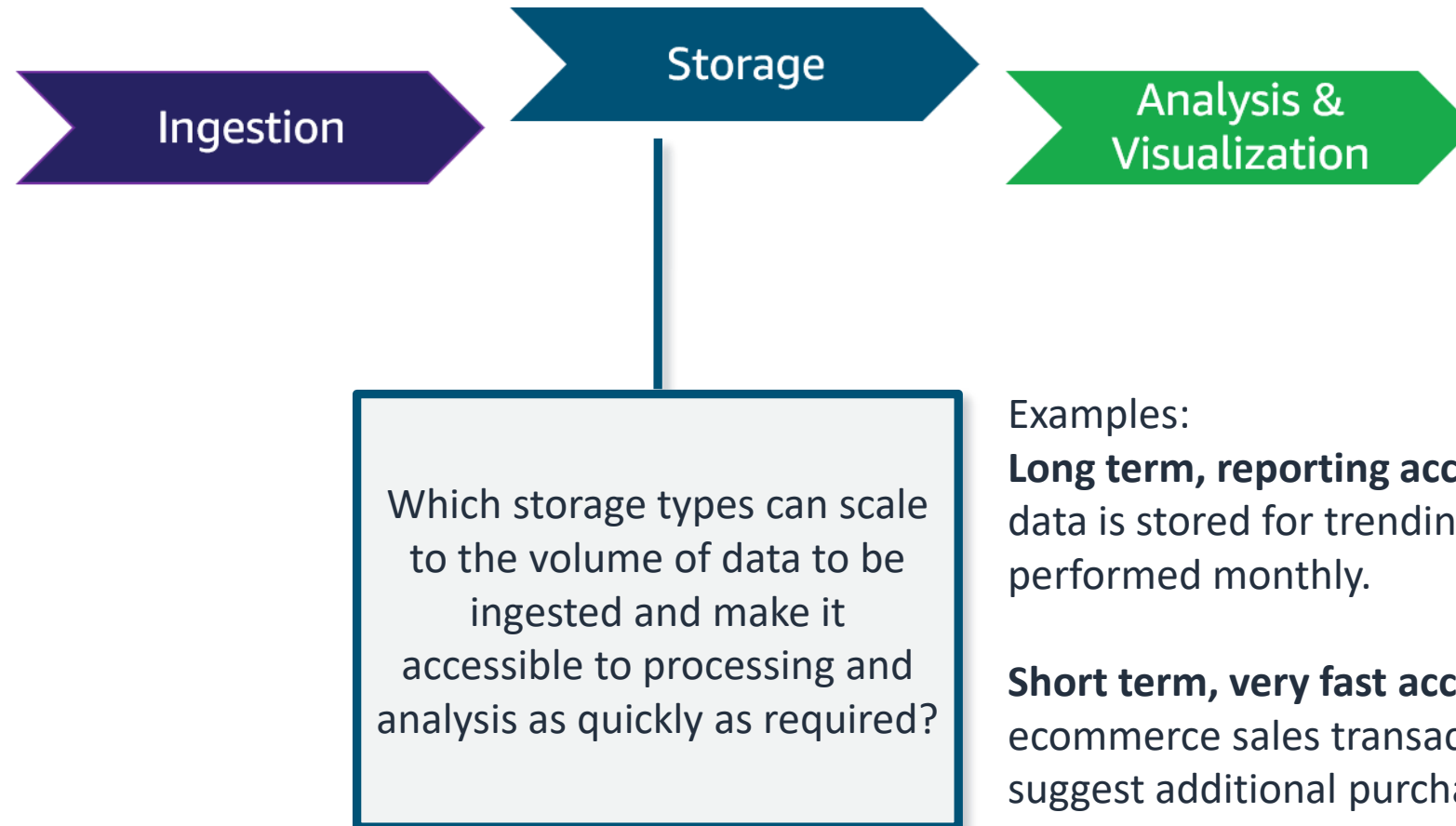
Which ingestion method suits both the amount of data to be ingested and the frequency with which new data must be ingested and processed?

Examples:

Streaming ingestion: Clickstream data from a retailer's website sends a large volume of small bits of data at a continuous pace. Analysis must be performed immediately.

Batch ingestion: Sales transaction data from retailers across the world is sent to a central location periodically. Data is analyzed overnight with reports sent to branches in the morning.

Storage decisions related to volume and velocity



Examples:

Long term, reporting access: Five years of sales data is stored for trending analysis, which is performed monthly.

Short term, very fast access: Incoming ecommerce sales transaction data is used to suggest additional purchases within the current session.

Processing decisions related to volume and velocity



How much data must be processed in a single iteration?
Does it require a distributed solution?
How quickly and how often does processing need to occur?

Examples:

Big data processing: Analytics is performed on credit card data for all US-based transactions in the past week.

Streaming analytics: Real-time alerts are produced on log data to identify potential fraud as soon as it occurs.

Analysis/visualization decisions related to volume and velocity



Examples:

Historical analysis: A year's worth of sales data is visualized, and users can drill down by region and salesperson.

Streaming Internet of Things (IoT) data: The real-time error rates of sensors in a plant are visualized.

How much data must be used in a single visualization? Do users need to drill down into details?

How quickly do consumers need to see and act on the data?

Key takeaways: Volume and velocity



- Volume is about how much data you need to process.
- Velocity is about how quickly data enters and moves through your pipeline.
- Volume and velocity together drive the expected throughput and scaling requirements of your pipeline.
- You should evaluate the volume and velocity requirements for each layer in your pipeline.
- Balance costs and throughput requirements.

Variety – data types

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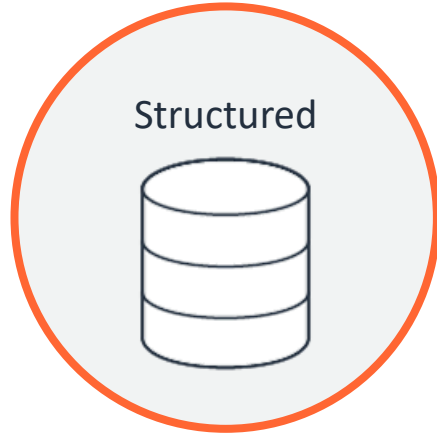
Designing your pipeline to support data variety

When deciding on the data that you will use, the variety of both data types and data sources will influence pipeline design on each layer of your pipeline.

- Different data types will lend themselves to certain types of processing and analysis.
- Different data source types might require different amounts of discovery and transformation work.
- The data source type will also drive the type and scope of the ingestion layer.

Data types

Easier to
use
"Hotter"

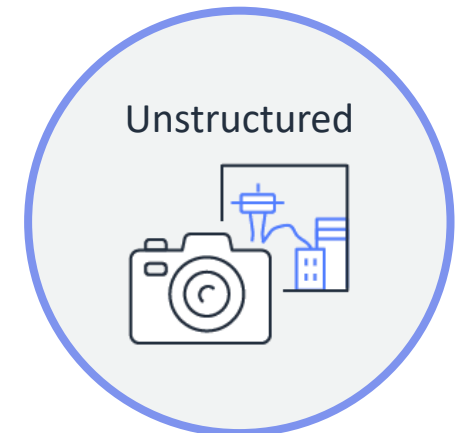


- Rows and columns
- Well-defined schema
- Example: Relational database tables



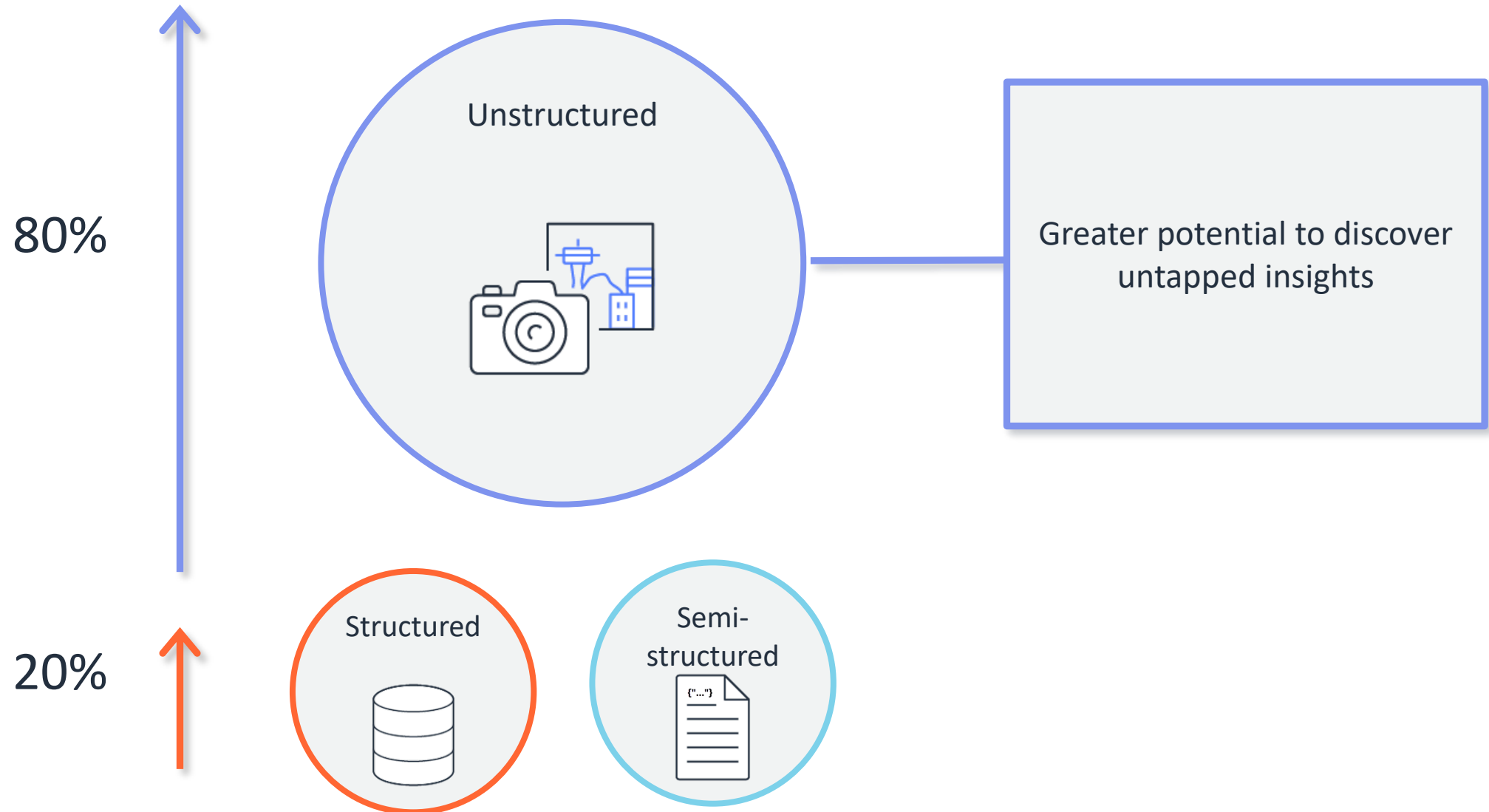
- Elements and attributes
- Self-describing structure
- Examples: CSV, JSON, XML

- Files
- No predefined structure
- Examples: Images, movies, clickstream data



More flexible
"Colder"

80% or more of available data is unstructured



Example use cases for data types

Structured

Query a relational database to report on customer service tickets that were submitted in a specific period.

Semistructured

Extract and analyze customer comments from an online chat application that saves conversations as JSON.

Unstructured

Perform sentiment analysis on customer service emails.

Key takeaways:

Variety – data types



- General data types include structured, semistructured, and unstructured.
- Structured data is the easiest to query but the least flexible.
- Unstructured data is the hardest to query but the most flexible.
- Most of the growth of available data is unstructured.

Variety – data sources

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Common data source types



On-premises databases or file stores

Application data is owned and managed by the organization.



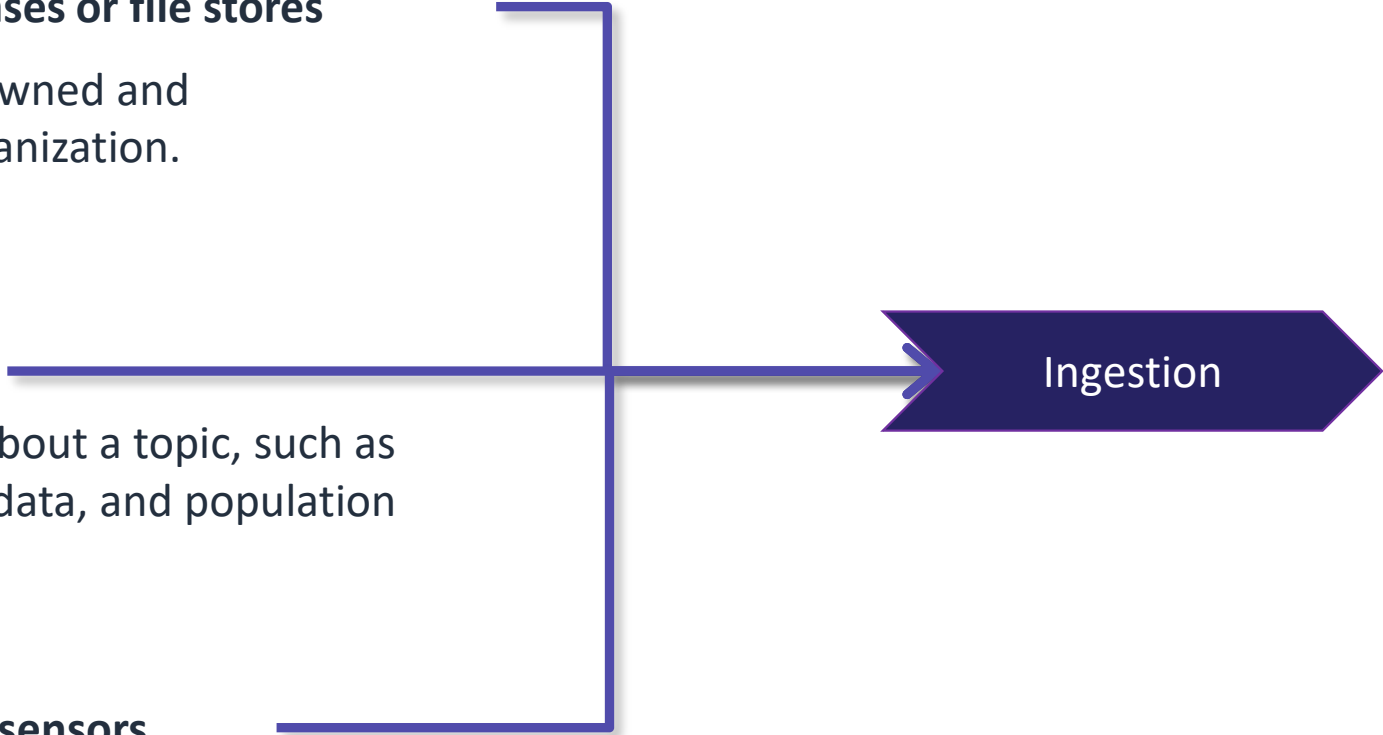
Public datasets

Data is aggregated about a topic, such as census data, health data, and population data.



Events, IoT devices, sensors

Data is generated continually by events and includes a time-based component.



Ingestion

Pipeline considerations based on data source type



On-premises databases or file stores

- Is controlled by the organization and might be ready for analysis
- Might contain private information
- Is often structured



Public datasets

- Contains data you might not need
- Probably requires transformation and merging with other data
- Is often semistructured data



Events, IoT devices, sensors

- Requires streaming ingestion and storage for time-series data
- Often requires real-time processing

Example use cases based on data sources



On-premises databases or file stores

A healthcare company runs analysis on customer data to identify patients who haven't received care lately.



Public datasets

A healthcare company combines public health data with customer data for a personalized mobile app that identifies demographic-based risk factors.



Events, IoT devices, sensors

A mobile app provides real-time heart rate monitoring and alerts when heart rate data coincides with a risk pattern.

The challenges of variety

- The way the data has been formatted and stored might impact your ability to analyze it.
- Ingestion and processing methods can become complex when you must combine different data types and sources.
- Data veracity can be more difficult to maintain when multiple data types and sources must be merged.

Key takeaways:

Variety – data sources



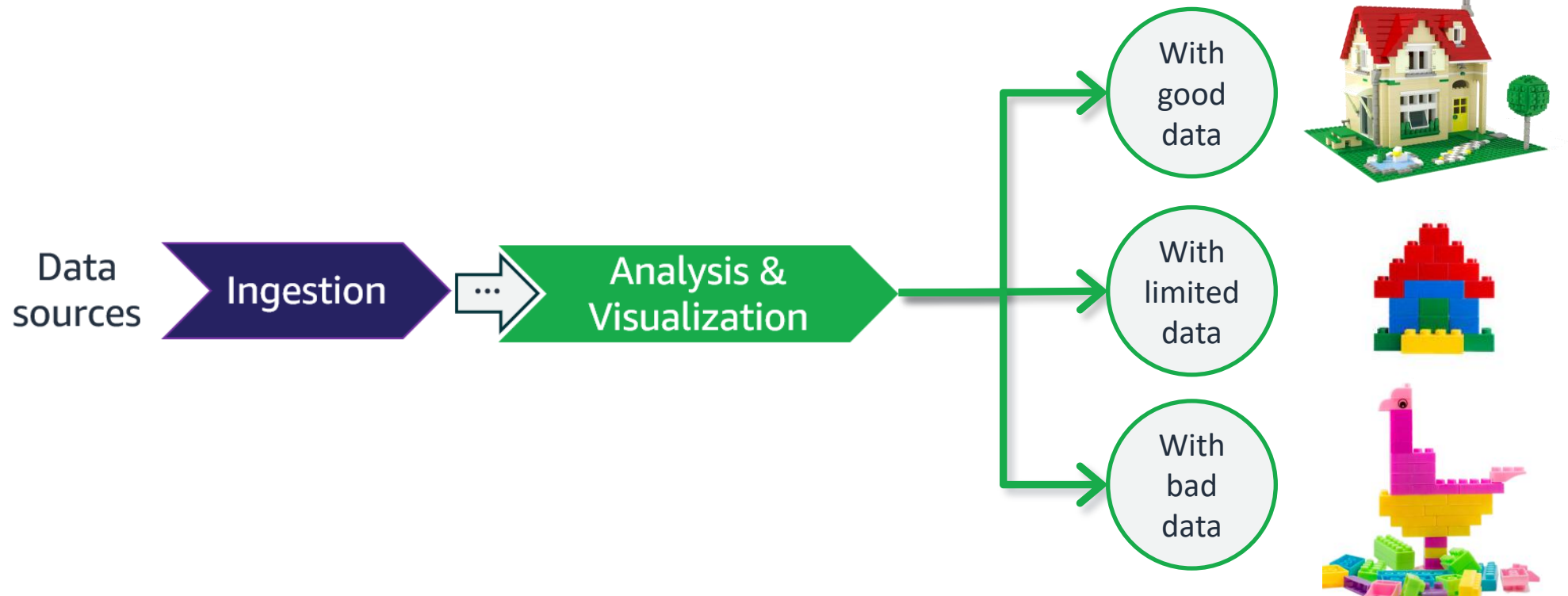
- Data source types include organizational stores, public datasets, and time-series data.
- Combining datasets can enrich analysis but can also complicate processing.

Veracity and value

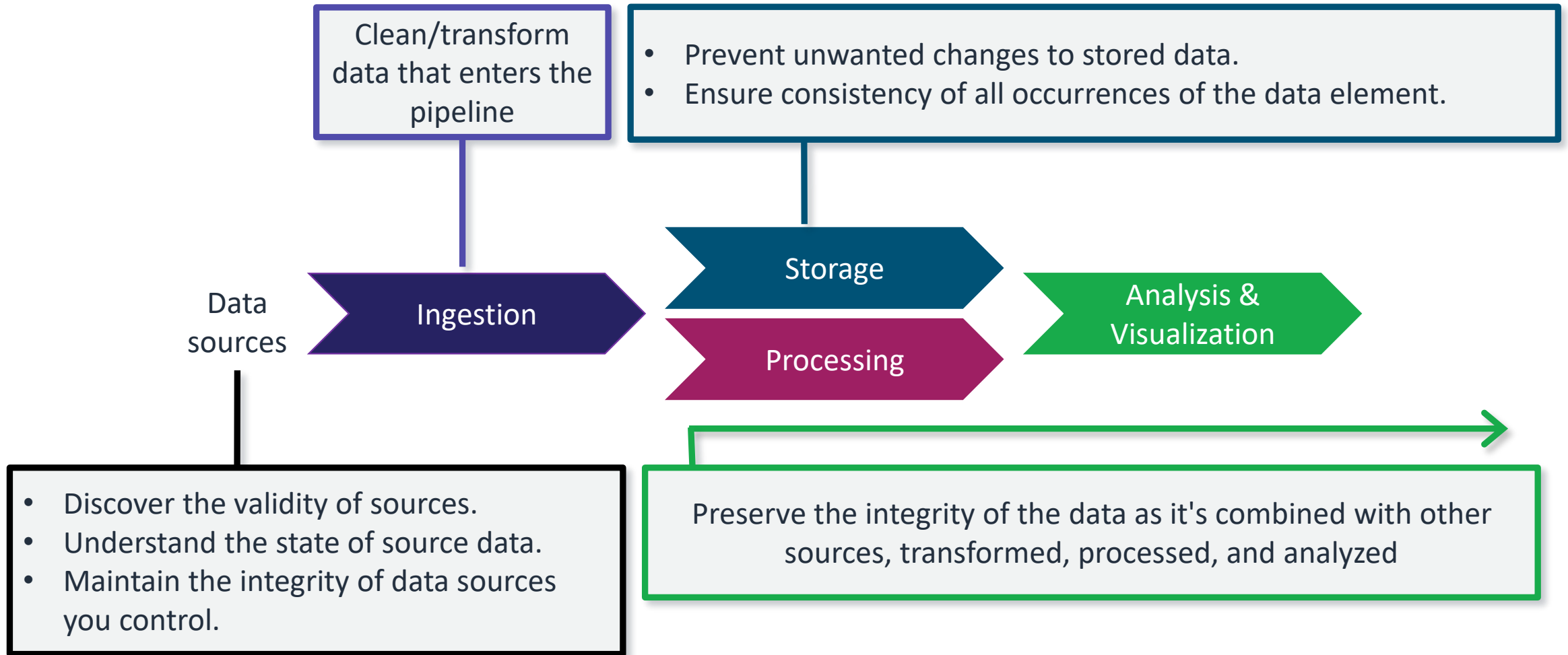
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Value rests on veracity

Making a data-driven decision with bad data is worse than making a decision without data.



Veracity across the pipeline



Examples of data issues that decrease veracity



Discover

Examples

- Dated information
- Missing data
- Lack of lineage
- Ambiguity
- Statistical bias



Clean/transform

Examples

- Duplicates
- Abnormalities
- Source differences



Prevent

Examples

- Software bugs
- Tampering
- Human error

Key takeaways: Veracity and value



- Veracity is about being able to trust the data that you are basing your analysis on.
- Value rests on veracity because without good data you could make bad decisions.
- You need to evaluate the veracity of each data source, clean and transform the data for your needs, and prevent unwanted changes to the data.

Activities to improve veracity and value

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Evaluating the veracity of data

Data engineering-focused questions:

- How is the source managed and maintained?
- Who generated the data, and who owns it?
- How often is the data updated?
- Do we need all fields and all records?
- Do audit trails exist?

Data science-focused questions:

- What methods were used to collect and process the data?
- Is this dataset the product of scientific research methods?
- What type of bias is likely part of this dataset?
- Does the data appear to be complete?
- How recent is the most recent data? Are there time gaps?



Best practices for cleaning data



Define clean

Come to agreement on what clean looks like for a source, and consistently apply the definition.

Trace errors

Don't just fix errors as part of your cleaning process—return to the source or sources and find the cause.

Change thoughtfully

Know what acceptable changes look like.
Ensure that cleaning processes don't make assumptions about what a data value means.

Retain auditable data

Don't discard raw data after cleaning if the raw data has business value.

Transformation example



Organization database record

Given name	Surname	Children	State
Paulo	Santos	<null>	Ohio

Transform

- <null> to 0
- state name to abbreviation

Ingestion

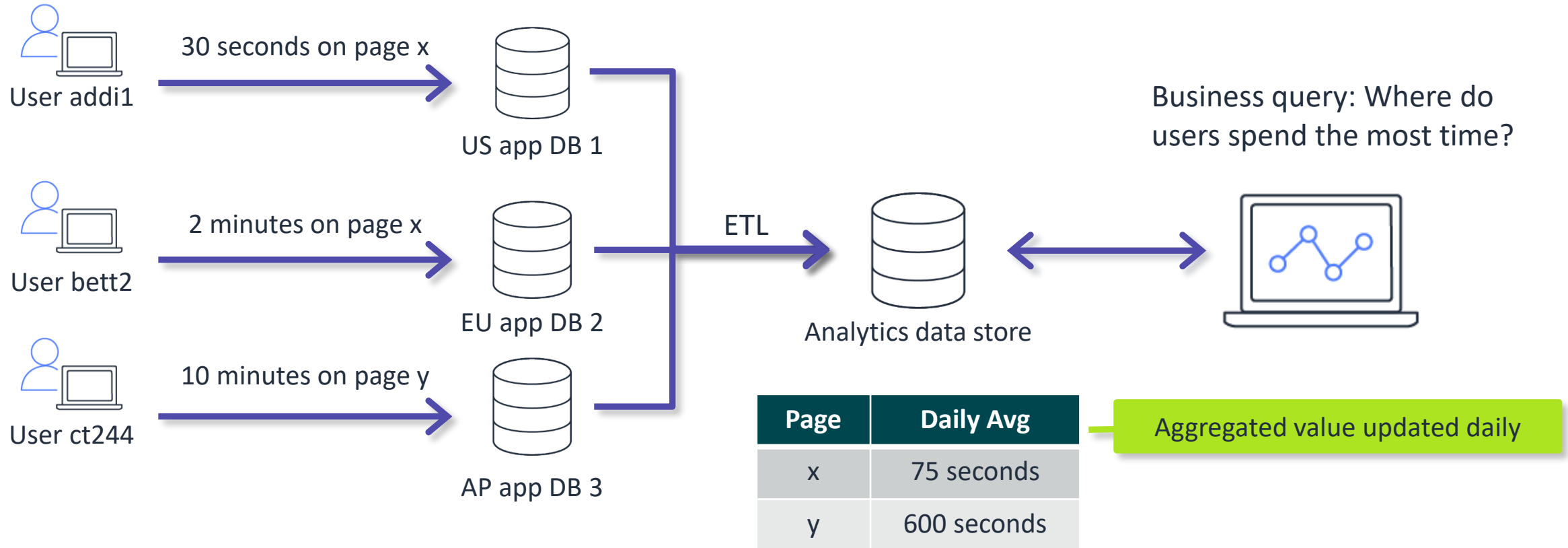


Public dataset record

```
{  
  "firstname": "John",  
  "lastname": "Doe",  
  "children": 3,  
  "State": "PA"  
}
```

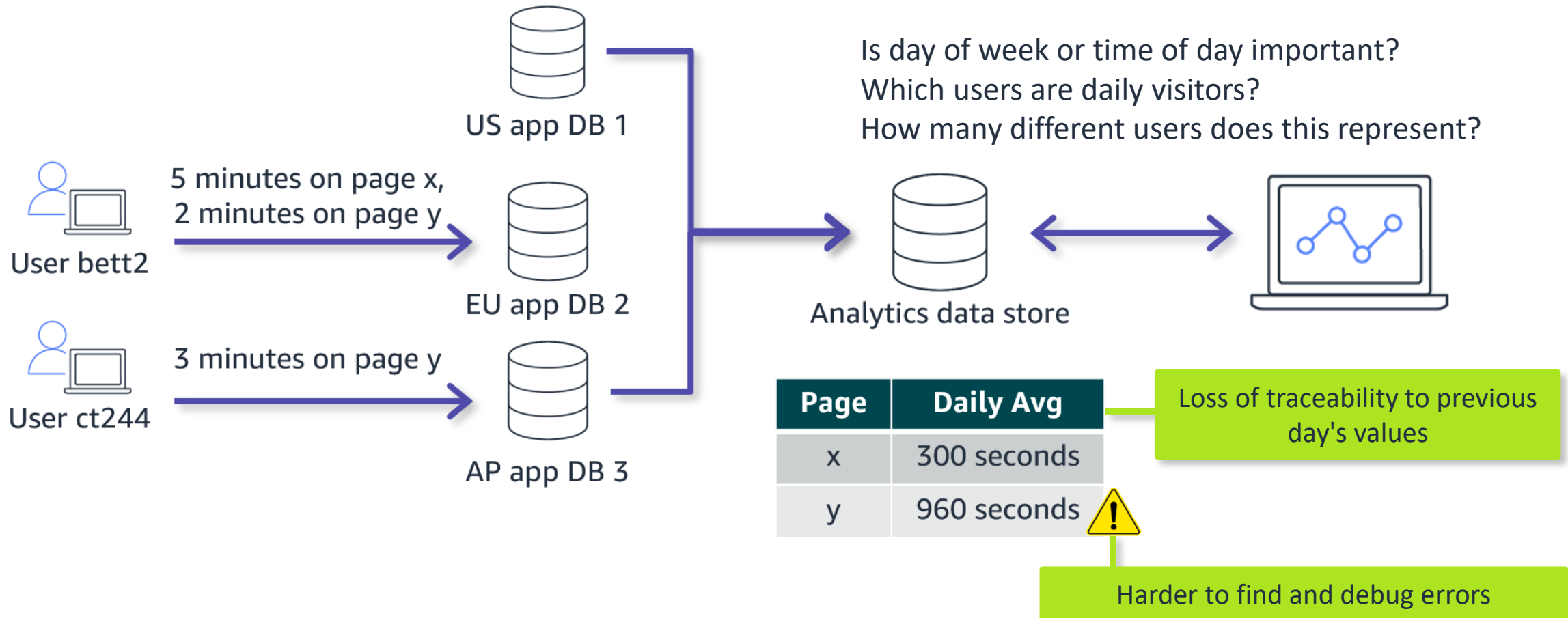
Saving aggregated data compared to saving raw data records

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Shortcomings of this approach for analytics

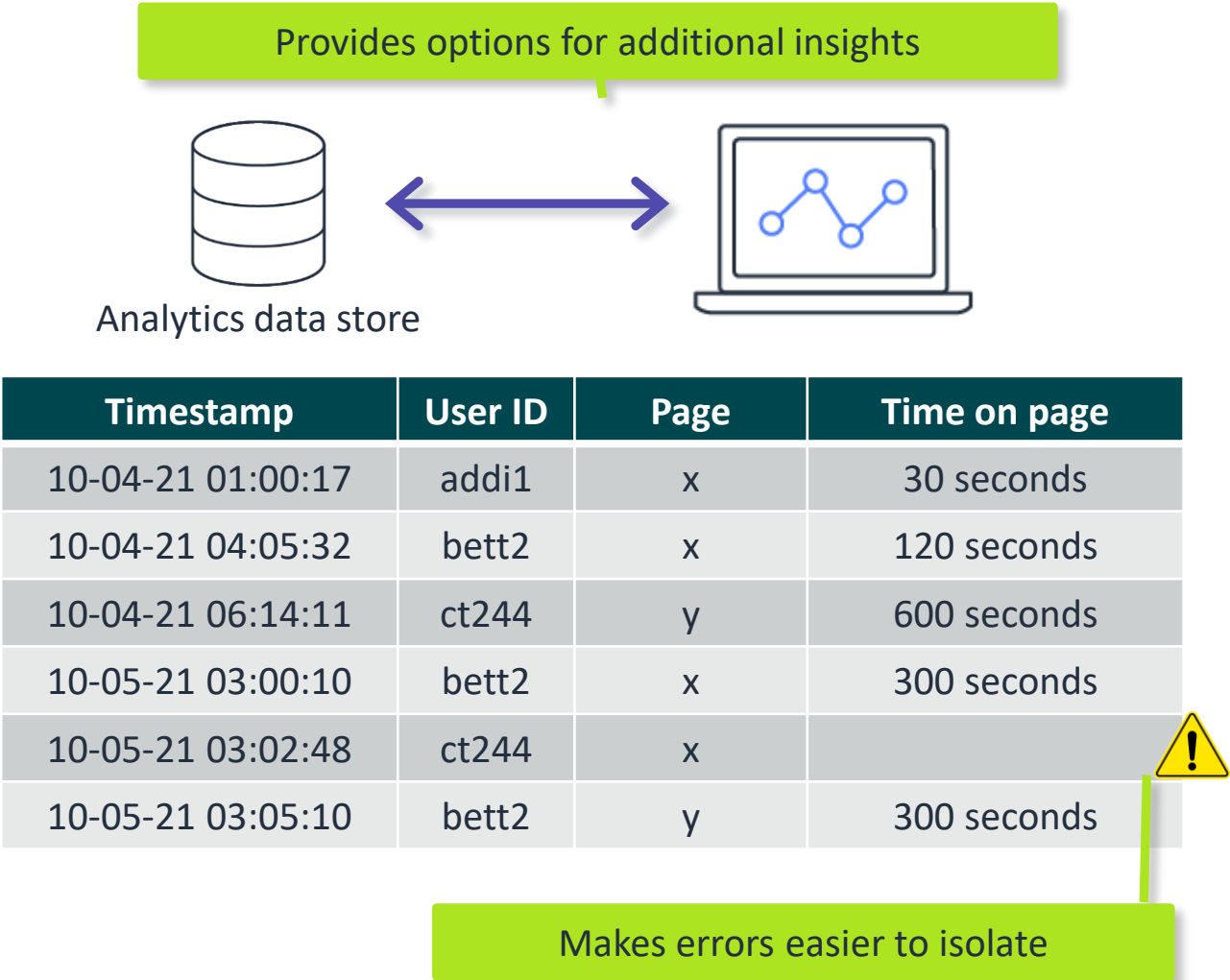
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Better approach for analytics: Immutable data

- Write a new record for each event.
- Timestamp each new record.
- Capture as much raw, unmanipulated data as you can.

Retains details and traceability to original values



Maintaining data integrity and consistency

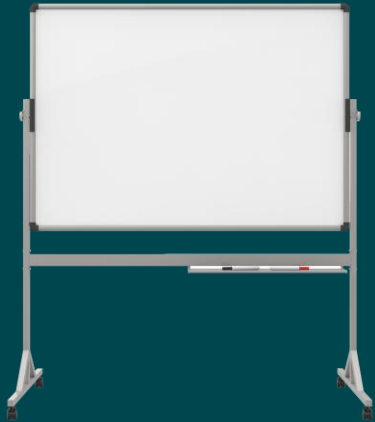
- Secure all layers of the pipeline.
- Grant least privilege access.
- Apply best practices to maintain data integrity.
- Keep audit trails.
- Implement data compliance and governance processes including data classification and data cataloging.
- Maintain a single source of truth.

Key takeaways: Activities to improve veracity and value



- You need to ask questions about the trustworthiness and lineage of the source data.
- To clean data, you need a common definition and must avoid making assumptions about what values should be.
- Data transformations can be simple, such as value substitutions, or complex, such as deriving new values.
- Save timestamped details instead of aggregated values.
- The organization should implement compliance and governance strategies to protect the integrity of data in your systems.

Activity: Planning Your Pipeline



- In this activity, you will select a scenario and document your ideas about planning a pipeline for the scenario. Use the 5 Vs of data to guide you.
- Use the detailed instructions and the worksheet provided in your online course to complete this activity.

Module wrap-up

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Module summary

This module prepared you to do the following:

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Module knowledge check



- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions that are based on material from the slides and slide notes.
- You can retake the knowledge check as many times as you would like.

Sample exam question

A data engineer has been asked to develop a pipeline to analyze partner sales trends by using sales transaction data. Each sales partner provides a .csv file each month. Although the files contain similar data, they are not in a standard format.

Which option is the most effective approach to collect this data for a data analytics pipeline?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- Using **sales transaction data**
- Monthly **.csv file** from **partners**
- Similar data, **not in a standard format**
- **Collect the data**

Sample exam question: Response choices

You have been asked to develop a pipeline to analyze partner sales trends by using **sales transaction data**. Each partner provides a **.csv file each month**. Although the files contain similar data, they are **not in a standard format**.

Which option is the most effective approach to collect this data for a data analytics pipeline?

Choice	Response
A	Extract the number of sales from each partner file. Then, write the partner ID and sales total to the analytics data store for analysis.
B	Extract the data from each partner's file, and transform it into a uniform file structure before writing it to the analytics data store for analysis.
C	Identify sources of statistical bias as part of extracting and transforming the data for analysis.
D	Import all of the files into the analytics data store with a partner ID. Grant access to each partner to edit the records and fix errors in only their data.

Sample exam question: Answer

The correct answer is B.

Choice	Response
B	Extract the data from each partner's file, and transform it into a uniform file structure before writing it to the analytics data store for analysis.

Thank you



Corrections, feedback, or other questions?

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